B · Before you begin

Read this chapter once before you pick up a drill. Budget a weekend: the steel on day one, the assembly on day two.

Safety

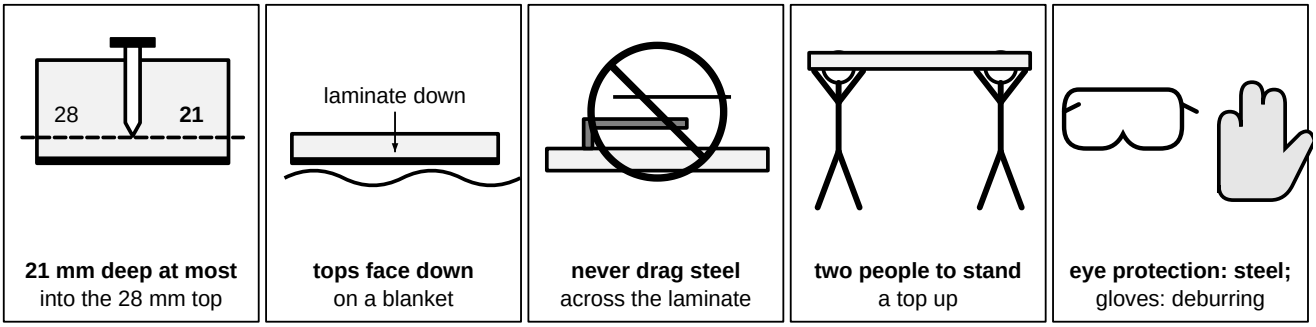
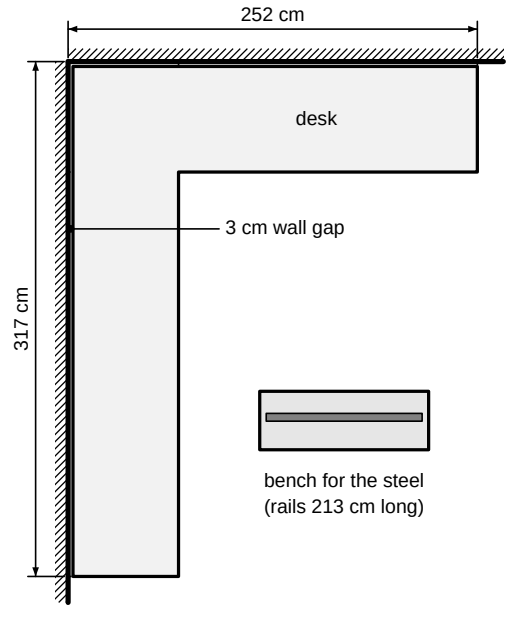


Fig. B-1 · The five rules that protect the tops and you.

> **NEVER A FASTENER DEEPER THAN 21 MM INTO A TOP**
> The EKBACKEN tops are 28 mm particleboard with a thin laminate skin. A screw or pilot that reaches the skin lifts it into a bump you will see and feel for ever. Every screw in this manual stops at least 7 mm short of the top face (28 − 21). If your tops measure 27 mm instead of 28, shorten every screw into the top by 1 mm. If you ever substitute a screw, keep this rule. Chapter D shows every depth.

- **Laminate protection.** Spread a blanket on the floor first. The tops lie face down on it, laminate on the blanket, for every marking and drilling step. Never slide or drag steel (rails, fittings, the clamp plates) across the laminate: lift it on and off.
- **Eye protection** for every hole drilled in steel; drill at low speed with cutting oil. **Gloves** while you deburr and file the steel: drilled edges are sharp.
- **Two people** to turn a top over and stand it on its legs (249 cm long). A helper is also needed to slide the finished desk off and back to the walls.

Work area



Plan, 1:40. Both walls MEASURE.

Fig. B-2 · Floor needed: 252 × 317 cm, including the 3 cm gaps to both walls.

The finished desk needs **252 × 317 cm** of clear floor in the corner: each top is 249 cm long and stands 3 cm off its wall. MEASURE ON HARDWARE Measure both walls (wall to wall) and the baseboard thickness before you start.

You also need floor space for one top at a time lying face down on the blanket (249 × 65 cm plus room to kneel all round), and a **bench** with two clamps for drilling the two 213 cm steel rails. All steel is drilled on the bench, never under the desk.

Clear the corner before chapter G: the ALEX units are placed by the walls before the tops go on.

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C · Complete parts inventory 11

D · Component preparation 14

Workshop tools

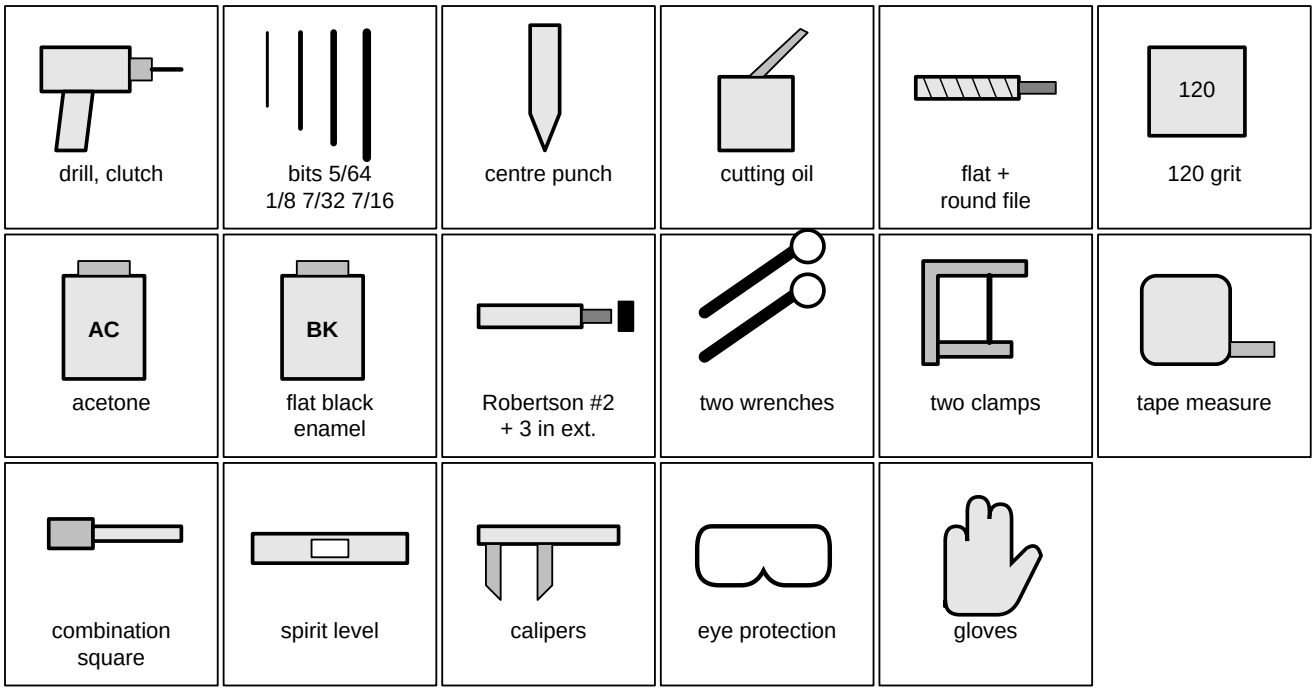


Fig. B-3 · Tools and protective equipment. Not to scale.

Tool	For
Corded or 18 V drill, clutch set low	all holes and screws
Bits: 5/64, 1/8, 7/32 in HSS; 7/16 in cobalt	ring and trim pilots; top and steel pilots; rail screw holes; fitting bolt holes
Centre punch (or a nail)	only for a mark the jig or plug cannot reach
Cutting oil or 3-in-1	every hole in steel
Flat file, round file, 120 grit, acetone, flat black enamel	deburr, ease a tight hole, scuff, degrease, paint the rails
Robertson #2 bit + 3 in magnetic extension	rail screws beside the vertical leg (the chuck would hit the leg otherwise)
Two wrenches or sockets for the 3/8-16 bolts (one for the head, one for the nut)	the corner fittings
Two clamps, 6 in or larger	hold each rail on its line; hold a fitting as a drilling template
Tape measure, combination square, masking tape, pencil	layout and the tape labels on the tops and rails
Spirit level, 60 cm or longer	setting the legs and feet
Calipers	top thickness, pad diameter, ALEX panel (the measurements on the last page of this chapter)

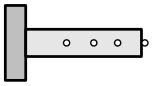
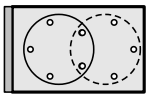
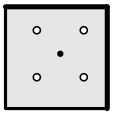
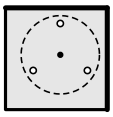
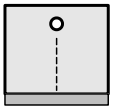
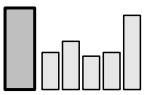
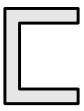
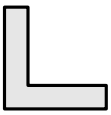
Protective equipment

- Safety glasses or goggles: every hole in steel, and opening the 7/16 in holes.
- Work gloves: deburring, filing and handling the cut rail ends.
- A blanket large enough for a whole top (249 × 65 cm).

Printed tools (P18): print these before you mark anything

All of them are in `parts/tools/`; `parts/README.md` has print settings. Print as saved, no supports. **Never scale a printed tool in the slicer: its distances are the tool.** The lipped and fenced tools print lip-up and are turned over to use.

> **EVERY JIG HOLE IS BIT-SIZED**
> Hold the jig on, start each pilot 3–4 mm through the jig's hole, lift the jig, then finish the hole to depth with the drill guide's puck. The small centre holes are for a scriber only; nothing is drilled through them.

Tool	What it does	Qty
	**Edge T-square.** Lip over any edge; holes named for their distance from the edge in mm: 50 (ring centres on the ALEX, 5 cm in), 85 (feet, 8.5 cm in, with the plate's four 1/8 in holes 46.8 mm apart round it), 100 (legs L2–L6, 10 cm in) and 150 (leg L1, 15 cm in).	1
	**Leg jig.** Lip over the wall edge; an OLOV's centre and its five 1/8 in holes in one placement: 100 pattern for L2–L6 (solid circle), 150 pattern for L1 (dashed circle, ringed holes).	1
	**Feet corner jig.** Fences on a far corner of a face-down top; a foot's centre 8.5 cm from both edges and its four 1/8 in holes 46.8 mm apart: F2, F4, F7, F8.	1
	**ALEX corner template.** Fences on an ALEX corner; ring centre 5 cm from both edges, the ring's three 5/64 in holes (radius 33 mm) and the three optional ALEX-foot holes (radius 14.5 mm). Top and bottom corners.	1
	**Rail marking gauge.** Lip over the angle's edge; pencil notches at 19 mm (screw line on the flat leg) and 29 mm (bolt line on the vertical leg).	1
	**Rail drill jig.** Lip over the flat leg's free edge; a 1/8 in bore on the 19 line; sight grooves on the station line. PETG.	1
	**Slot plug** (fitting-centring-plug). Drops into a fitting's 21 × 12 slot; a 1/8 in bore dead centre, so the pilot cannot wander in the slot's play.	2
	**Pilot drill guide.** SET tower (45 mm for the 1/8 in bit, 35 mm for the 5/64 in) and five depth pucks: 12 PLATE, 15 LEG, 10 EDGE, 12 ALEX, 25 RAIL. Re-set the bit every time you change it.	1 set
	**Foot height gauge.** A C with 102 mm inside: all eight feet wound to the same height without a ruler.	1
	**Wall gauge.** An L: the 65 leg (6.5 cm, ALEX back to wall) and the 30 leg (3 cm, a top's raw edge to wall).	1

Also print the eight locating rings (P10) now: `parts/structure/foot-ring`, PETG or ABS, flange down. Their 51 mm bore depends on the pad diameter you measure (last page of this chapter).

Printed tools: how each one is used

1 · Edge T-square

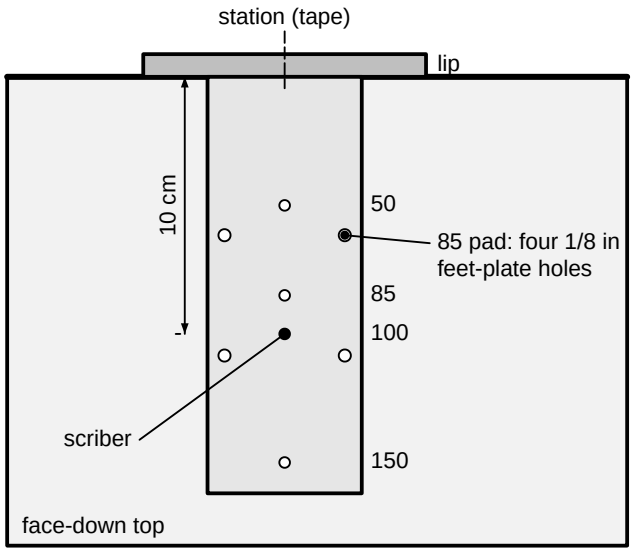


Fig. B-4 · The T-square on a face-down top, lip over the edge, at its station. Hole names are mm from the edge.

1. Turn it over so the lip hangs down. Hook the lip over the edge named in the marking table.
2. Slide the head along the edge until the blade's centre line sits on the station you ticked with the tape.
3. Put the scriber through the hole the table names: **50** ring centres on the ALEX (5 cm in), **85** feet (8.5 cm in), **100** legs L2–L6 (10 cm in), **150** leg L1 (15 cm in). Each hole is named for its distance from the edge in mm.
4. The four holes round **85** are 1/8 in drill holes for a feet plate (46.8 mm apart): start the four pilots through them, lift, finish with the 15 LEG puck.
5. **First-print check:** hook it over a top's edge: the 100 hole must sit 10.0 cm from the edge. If not, re-print it, never scaled.

2 · Leg jig

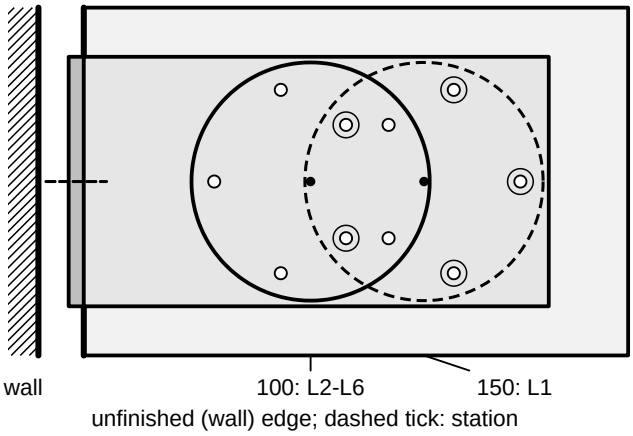


Fig. B-5 · The leg jig hooked over the wall (unfinished) edge.

1. Turn it over (lip down). Hook the lip over the wall edge and slide it to the leg's station.
2. **L2–L6:** the **100** pattern: plain holes, solid circle, one hole toward the wall. **L1:** the **150** pattern: ringed holes, dashed circle.
3. Start the five 1/8 in pilots 3–4 mm deep through the pattern's holes.
4. The small centre hole is for a scriber only; never drill it.
5. Lift the jig and finish each pilot with the **15 LEG** puck.

3 · Feet corner jig

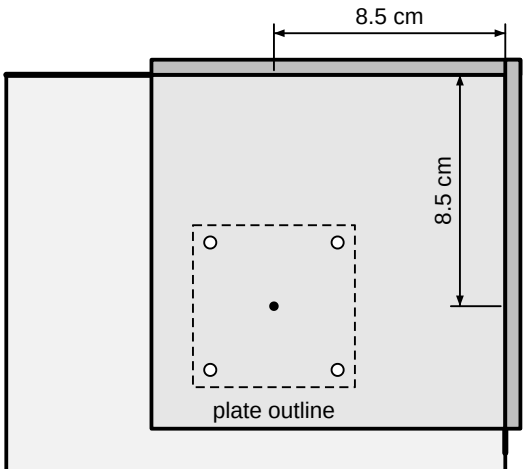


Fig. B-6 · The feet corner jig on a far corner of a face-down top, fences against both edges.

1. Turn it over (fences down). Drop it over a **far corner** of the top so both fences bear on the two edges.
2. It gives the foot centre 8.5 cm from both edges and the plate's four holes, 46.8 mm apart, square to the edges.
3. Start the four 1/8 in pilots through it, lift it, finish with the **15 LEG** puck.
4. It does **F2, F4, F7, F8**. The other four feet (F1, F3, F5, F6) are 34.5 cm from the end: the T-square's 85 pad does those, at the tape stations in chapter D.

4 · ALEX corner template

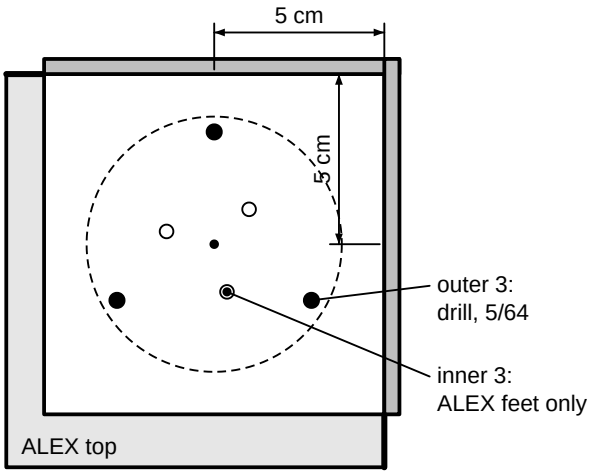


Fig. B-7 · The template on an ALEX top corner. Black: the three ring holes you drill. White: the ALEX-foot holes.

1. Turn it over (fences down). Drop it over a corner of a cabinet's **top**, fences against both edges: ring centre 5 cm from both edges.
2. Start the **three outer** pilots (5/64 in, radius 33 mm) through it. Lift it, finish with the **12 ALEX** puck.
3. The **inner three** (radius 14.5 mm) are only for the optional ALEX feet, marked on the **bottom** corners. Do not drill them on the top.
4. Set the ring on the corner and turn it until its three countersunk holes sit on the three pilots; it only fits one way round. Same template for all eight corners.

5 · Rail marking gauge

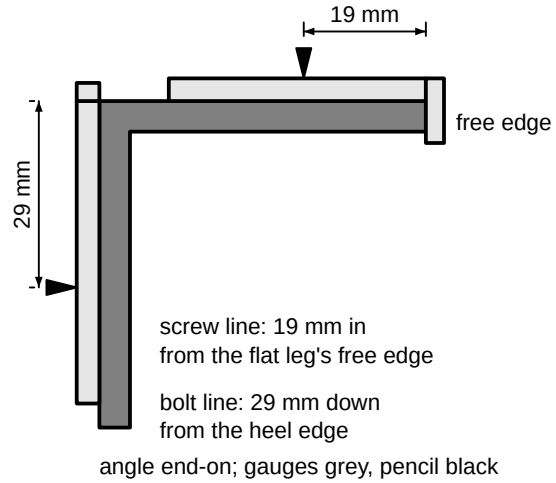


Fig. B-8 · The angle end-on with the gauge in its two positions.

1. Turn it over (lip down). Lay it on a face of the angle with the lip over an edge; the lip's inner face is the zero.
2. **Screw line:** lip over the flat leg's free edge, pencil in the **19** mm notch, slide the full length.
3. **Bolt line:** lip over the vertical leg's heel edge, pencil in the **29** mm notch, slide.
4. Use a 0.8 mm pencil point: the notches put it on exactly 19.0 and 29.0 mm.

6 · Rail drill jig

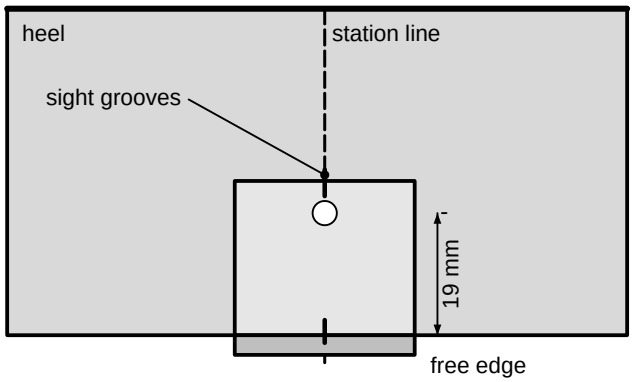


Fig. B-9 · The rail drill jig on the flat leg (seen from above), lip under the free edge.

1. Turn it over (lip down). Hook the lip over the flat leg's free edge.
2. Slide it until the sight grooves on both sides sit on the station line.
3. Hold it down and drill the 1/8 in pilot through the bore: it is on the 19 mm screw line, square, and cannot wander.
4. Pull the jig off. Open the hole to 7/32 in without the jig; the bit follows the pilot. Nineteen holes, one jig.

7 · Slot plug (fitting-centring-plug)

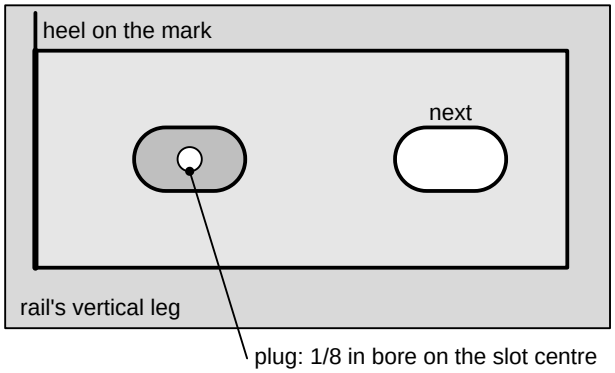


Fig. B-10 · A fitting clamped on the rail's vertical leg, heel on its mark, plug in the first slot.

1. Clamp the fitting to the rail with its heel on the mark from chapter E.
2. Drop a plug into the first slot. Drill the 1/8 in pilot through its bore. Move the plug to the next slot and repeat.
3. Take the fitting off. Open each pilot to 7/16 in; the bit follows the pilot.
4. **Why:** the slots are 21 × 12 mm, so a bolt or a punch could sit anywhere along the slot. The plug fills it and puts every pilot on the slot centre.

8 · Pilot drill guide

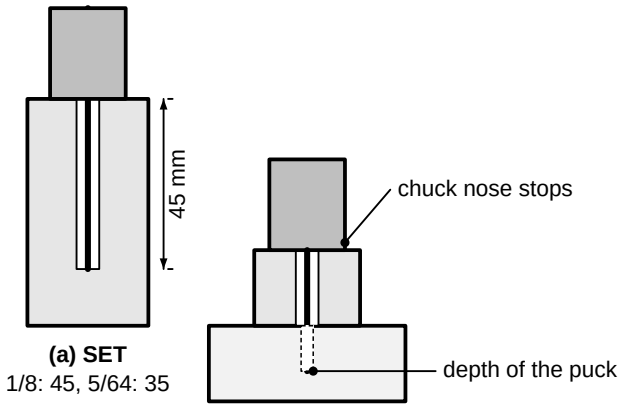
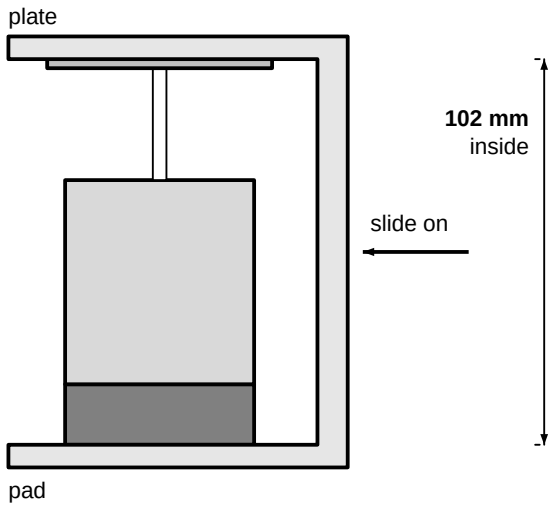


Fig. B-11 · (a) Setting the bit in the SET tower. (b) Drilling through a puck.

1. **Set the bit:** loosen the chuck, drop the bit tip-first into the SET tower's hole for it (45 mm for 1/8 in, 35 mm for 5/64 in), rest the chuck nose on the tower, tighten.
2. **Drill:** stand the puck for that hole flat on the surface (or on the plate, or on the rail's flat leg), bit down its bore, drill until the chuck nose lands on the puck.
3. **Pucks:** the number is the hole depth in mm. 12 PLATE (1/8 in, joining plates) · 15 LEG (1/8 in, OLOV and feet plates) · 10 EDGE (5/64 in, trim screws sideways into the raw edge, bit horizontal) · 12 ALEX (5/64 in, rings and ALEX feet) · 25 RAIL (1/8 in, through the angle, 20 mm into the wood).
4. Re-set the bit every time you change it. A domed chuck nose: drill one test hole in scrap and measure it.

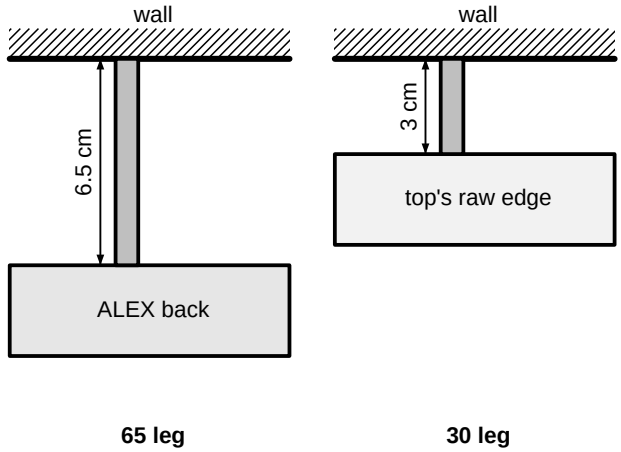
9 · Foot height gauge



1. Stand the foot on its pad.
2. Slide the C over it from the side.
3. Wind the pad until the gauge just slides on with no rock: 102.0 mm inside.
4. All eight feet are then identical, no ruler.

Fig. B-12 · The C gauge on a round foot: plate face to pad bottom.

10 · Wall gauge



1. Hold the **65** leg (6.5 cm) between the wall and a cabinet's back while you place it, and again when you slide the desk back after the trim.
2. Hold the **30** leg (3 cm) between the wall and a top's raw edge to check the 3 cm gap.
3. No tape needed: it just fits, or it does not.

Fig. B-13 · The two legs of the wall gauge in the gap.

THE RULE FOR EVERY PRINTED TOOL
Every jig hole is bit-sized: start the pilot 3–4 mm through the jig, lift the jig, finish the hole to depth with the puck. Centre holes are for a scriber only. Never scale a tool in the slicer: its distances are the tool.

How sure is each number?

Every dimension in this manual carries one of three confidence labels, or a fourth label that is an instruction to you:

VERIFIED	Measured on the purchased part, or taken from the part's own geometry file (the printed parts).
SOURCE-DEFINED	Defined by the project documents (the build manual): a design position, not a measurement. It is right by definition; you put the part there.
ASSUMPTION / FIELD CHECK	An engineering estimate or a check done on site. Confirm it before you rely on it.
MEASURE ON HARDWARE	Not yet measured on your parts. Measure it yourself before the step that uses it, and stop if it differs.

Confirm before you drill anything

> **MEASURE BEFORE DRILLING**
> These numbers are not yet confirmed on the parts you have. Each one is checked again where it is used; if one differs, stop and correct the layout before any hole is drilled.

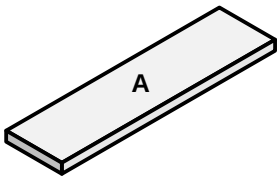
Measure	Expected	Label	Used in
EKBACKEN thickness, both tops (calipers)	28 mm	MEASURE ON HARDWARE	D: every screw depth into the tops. At 27 mm, shorten every screw into the top by 1 mm.
Delivered rail lengths, both pieces, and the angle thickness	213 cm layout length (bought as 84 in = 213.4 cm); 4.8 mm thick	MEASURE ON HARDWARE	E: hole stations. 213 cm is the layout length; the far end may carry the extra.
Strut fitting slots, positions from the heel	21 × 12 mm slots, centres 29.15 and 78.45 mm from the heel (legs 100.5 × 41 mm)	VERIFIED	E, F: measured 23 Oct 2026; the fitting with the slot plug is still the drilling template.
Feet pad diameter (calipers)	50 mm	MEASURE ON HARDWARE	G: the rings are bored 51 mm for a 50 mm pad; re-print them if yours differs.
ALEX top panel thickness, at an edge	16 mm or more for the 5/8 in screws	MEASURE ON HARDWARE	G: ring screws, 13 mm into the ALEX top.
ALEX height, with and without its stock feet	70.0 cm	ASSUMPTION / FIELD CHECK	G: with the feet it sets the underside at 80.2 cm.
VIVO top arm reach; maximum clamp opening	reach about 89 mm ASSUMPTION / FIELD CHECK; opening 31 mm or more	MEASURE ON HARDWARE	I: the clamps on the 28 mm top plus a pad plate.
Room: wall to wall along both walls; baseboard thickness	252 cm and 317 cm	MEASURE ON HARDWARE	B (work area), G (placing the ALEX units by the walls).
Feet load rating	unknown; the design wants 50 kg or more each	ASSUMPTION / FIELD CHECK	C (P09); check the listing or the maker.

Already measured and printed as VERIFIED: the OLOV plate (ø105 mm, five 5.5 mm holes on an 85 mm circle), the feet plate (59.5 mm, four 5.2 mm holes 46.8 mm apart), the strut fittings (21 × 12 mm slots at 29.15 and 78.45 mm from the heel), the VIVO crossbar (50.2 cm centre to centre), pole (35 mm), pole inset (43.7 mm), clamp back (80 × 17.5 mm) and lower pad.

C · Complete parts inventory

Check every part against this list before you start. Part numbers P01–P21 and hardware numbers H01–H11 are used in every callout balloon in this manual. Lengths a tape measures are in cm; thicknesses, holes, plates and screws in mm. REQUIRED parts are needed for the desk to stand; OPTIONAL parts can be left out.

thickness drawn ×4



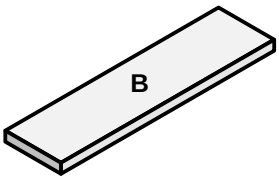
P01 **× 1**

Top A, IKEA EKBACKEN brown walnut, 249 × 65 cm, 28 mm thick. Front long edge rounded and laminated; back long edge square and unfinished (wall side).

MEASURE ON HARDWARE thickness

REQUIRED

thickness drawn ×4

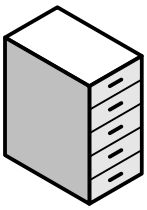


P02 **× 1**

Top B, identical to P01: 249 × 65 cm, 28 mm thick. Becomes B by being laid along the side wall (chapter D).

MEASURE ON HARDWARE thickness

REQUIRED



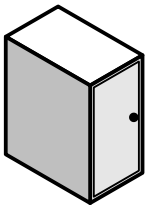
front

P03 **× 1**

ALEX 1, IKEA drawer unit, 36 × 58 × 70 cm. Under top A's far end, drawers toward you.

ASSUMPTION / FIELD CHECK height 70 cm

REQUIRED



front

P04 **× 1**

ALEX 2, IKEA storage unit with door, 36 × 58 × 70 cm. Under top B's far end, door toward you.

ASSUMPTION / FIELD CHECK height 70 cm

REQUIRED

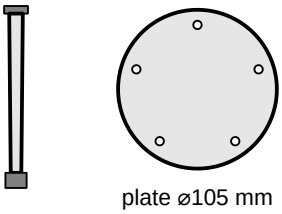
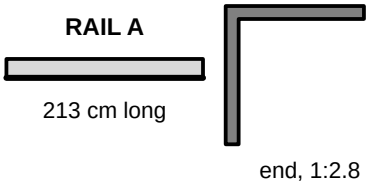


plate ⌀105 mm

P05 **× 6**

IKEA OLOV leg, adjustable 60–90 cm, set to 80.2 cm. Plate ⌀105 mm, five ⌀5.5 mm holes on an ⌀85 mm circle. Max 13 kg each. VERIFIED plate

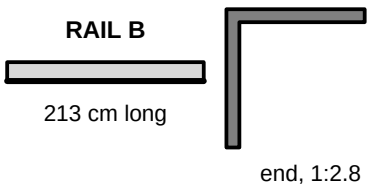
REQUIRED



P06 **× 1**

Rail A, 2 × 2 × 3/16 in hot-rolled steel angle (50.8 × 50.8 × 4.76 mm), 213 cm layout length (bought as 84 in = 213.4 cm). MEASURE ON HARDWARE length

REQUIRED

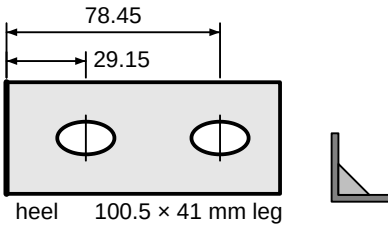


P07 **× 1**

Rail B, identical to P06 until its corner end is taped (chapter D).

MEASURE ON HARDWARE length

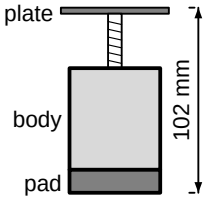
REQUIRED



P08 **× 2**

Gusseted 90° strut fitting, 5 mm steel, for 1-5/8 in channel. Legs 100.5 × 41 mm; two 21 × 12 mm slots per leg, centres 29.15 and 78.45 mm from the heel. VERIFIED

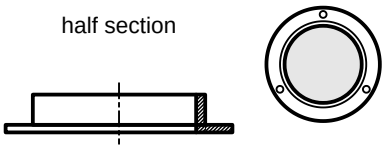
REQUIRED



P09 **× 8 (+4)**

Round adjustable foot, 4 in, wound to the 102 mm gauge. Body and pad ⌀50 mm; plate 59.5 mm square, four ⌀5.2 mm holes 46.8 mm apart. 12 on hand: 8 used, 4 SPARE. VERIFIED plate · MEASURE ON HARDWARE pad · load ASSUMPTION / FIELD CHECK

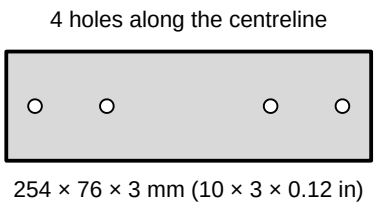
REQUIRED



P10 **× 8**

Locating ring, printed PETG or ABS: bore 51 mm, OD 57 mm, ring 10 mm high on a ⌀75 × 2.5 mm flange, three countersunk ⌀3.8 mm holes on radius 33 mm. **VERIFIED** geometry file

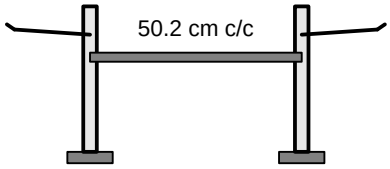
REQUIRED



P11 **× 4**

Mending plate, 10 × 3 × 0.12 in (254 × 76 × 3 mm), holes at 2.0, 7.0, 18.4, 23.4 cm from one end. Two join the tops (P1, P2); two pad the VIVO clamp jaws. **SOURCE-DEFINED**

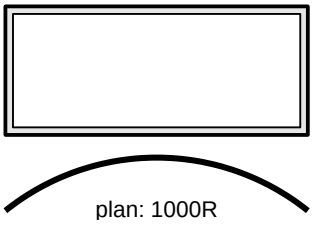
REQUIRED



P12 **× 1**

VIVO STAND-V300CU: two clamps, two ⌀35 mm poles 61 cm (24 in) tall, crossbar 50.2 cm centre to centre, two side arms, centre bracket. Clamp back 80 mm along the edge, 17.5 mm behind it. **VERIFIED**

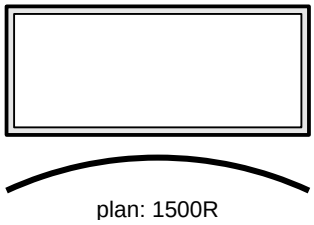
REQUIRED



P13 **× 1**

Samsung Odyssey G5 32 in, 1000R, VESA 75, about 5 kg. Centre screen, landscape, bottom edge 23 cm above the top. **SOURCE-DEFINED**

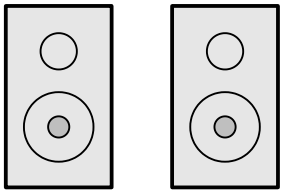
REQUIRED



P14 **× 2**

Acer XZ320QU 31.5 in, 1500R, VESA 100, 5.6 kg with stand (about 5 kg on the arm). Side screens, landscape. **SOURCE-DEFINED**

REQUIRED



P15 **× 2**

Edifier R1280DBs speakers, 23.4 cm tall, one under each Acer. **SOURCE-DEFINED**

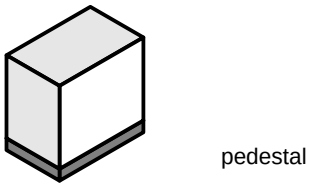
REQUIRED



P16 **× 1**

IKEA ELLOVEN monitor stand with drawer, 10 cm tall, under the G5. **SOURCE-DEFINED**

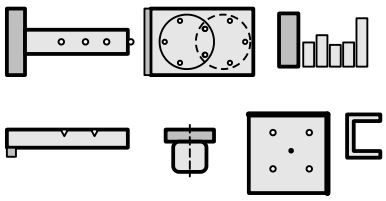
REQUIRED



P17 **× 1**

HYTE Y60 on its printed pedestal (MakerWorld 679012): footprint 49 × 31 cm, 48 cm tall with the pedestal, over ALEX 1. **SOURCE-DEFINED**

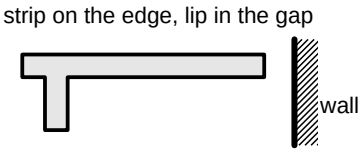
REQUIRED



P18 **1 set**

Printed tools: T-square, leg jig, feet corner jig, ALEX corner template, rail gauge, rail drill jig, slot plug × 2, drill guide (tower + 5 pucks), foot height gauge, wall gauge. List in chapter B.

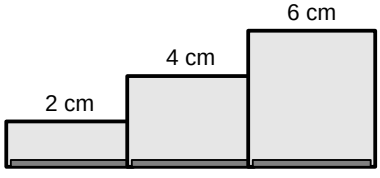
REQUIRED



P19 **1 set**

Wall-gap trim, printed: corner, fillers, clamp pieces, segments, connector tiles. Cut list in `parts/README.md`. Fixed with H07.

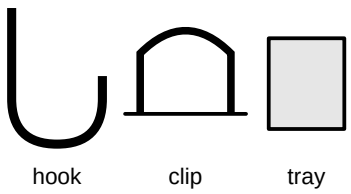
OPTIONAL



P20 **× 8**

ALEX feet, printed: ⌀56 mm, one height for all (2, 4 or 6 cm), with TPU pads. Raise the whole desk by their height.

OPTIONAL



P21 **1 set**

Cable parts, printed: rail cable hooks (magnets H11), cable clips, power-bar cradles, cable trays.

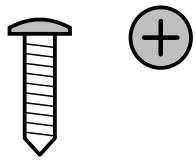
OPTIONAL

Drawings are silhouettes for recognition, not to scale unless a scale is written on them.

Hardware H01–H11

Screws and bolts are drawn **1:1** (true length, nominal diameter): lay a screw on the drawing to check it. Washers and the nut are symbols. All hardware was ordered 22 Oct 2026 SOURCE-DEFINED. Buy a few spares of each screw.

H01 × 30

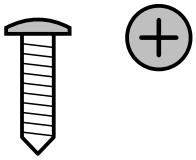


1:1 · head as supplied

M4 × 16 machine screw, fully threaded, IKEA 108443 (with the legs). OLOV plates, 5 per leg.

REQUIRED

H02 × 32

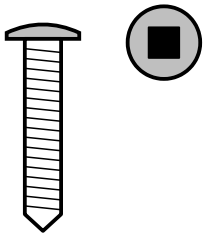


1:1 · pan head

#8 × 5/8 in pan head. Feet plates, 4 each.

REQUIRED

H03 × 19

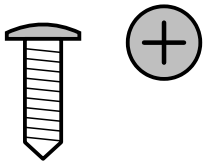


1:1 · pan head, Robertson

#10 × 1 in pan head, Robertson. Rails into the tops: 9 in rail A, 10 in rail B.

REQUIRED

H04 × 8

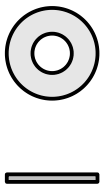


1:1 · pan head

#10 × 5/8 in pan head. Joining plates P1, P2, 4 each.

REQUIRED

H05 × 8

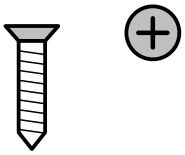


symbol, not to scale

#10 flat washer, under each H04.

REQUIRED

H06 × 24

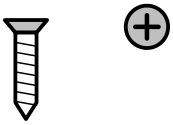


1:1 · flat head

#6 × 5/8 in flat head. Rings, 3 each. +24 for ALEX feet, more for cable parts.

REQUIRED

H07 × 48

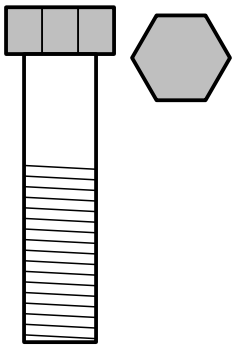


1:1 · flat head

#4 × 1/2 in flat head. Wall-gap trim, two per piece.

OPTIONAL

H08 × 6

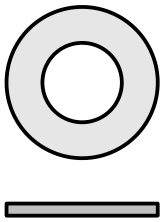


1:1 · hex head

3/8-16 × 1-1/2 in hex bolt, grade 5. Corner fittings.

REQUIRED

H09 × 12

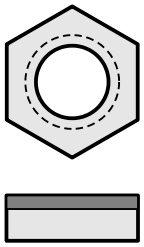


symbol, not to scale

3/8 flat washer, under each bolt head and each nut.

REQUIRED

H10 × 6

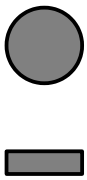


symbol, not to scale

3/8-16 nylon-insert lock nut. Stiff to turn; that is normal.

REQUIRED

H11 × 8



1:1

10 × 3 mm neodymium disc magnet, two per rail cable hook.

OPTIONAL

D · Component preparation

This chapter marks both tops face down and labels the two rails. Nothing is fixed yet. Positions are in **cm**, as the tape reads them; thicknesses, holes and screws in **mm**. Every position is measured from one **datum**: **top A's back-left corner**. **x** runs along the back wall, **y** along the side wall. Top A covers x 0–249 cm, y 0–65 cm; top B covers x 0–65 cm, y 65–314 cm.

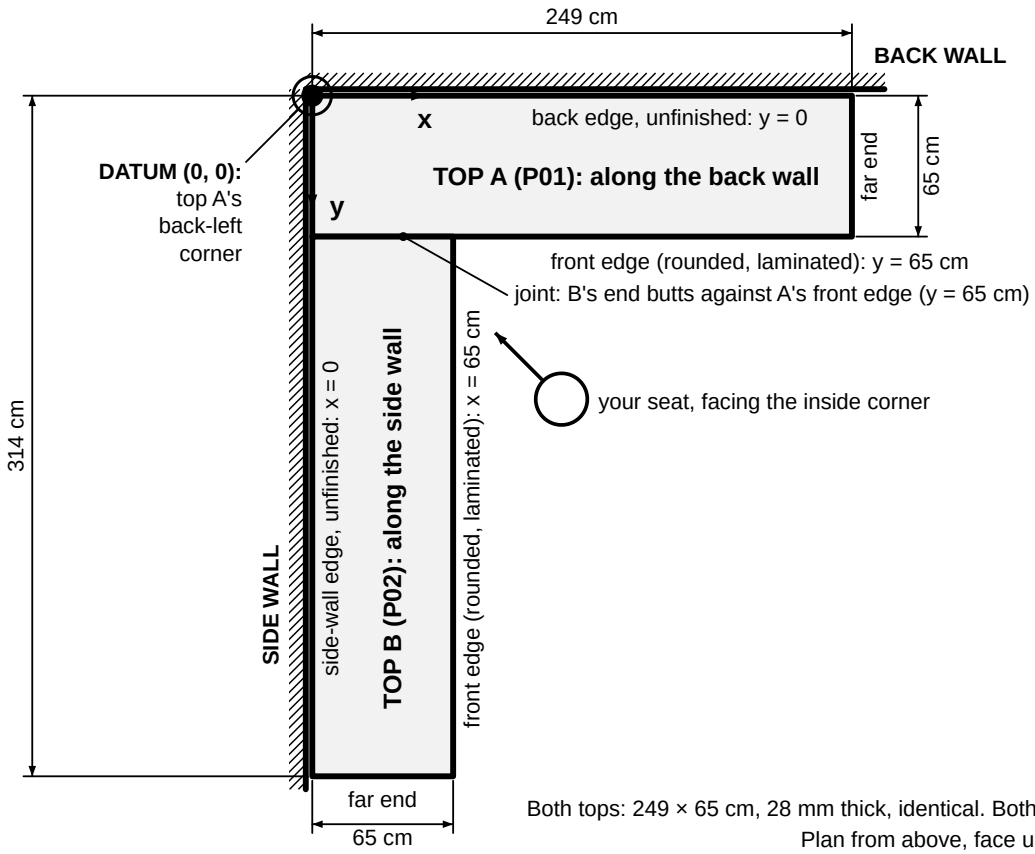


Fig. D-1 · The datum and the two tops, seen from above as installed. SOURCE-DEFINED

Which edge is which

Each top has one **rounded, laminated** long edge: the **front**. The other long edge is **square and unfinished**: it goes against the wall. Either short end can become the side-wall end.

Which top is A

The two tops (P01, P02) are identical. **A** is the one you lay along the back wall; its unfinished edge is y = 0. **B** lies along the side wall; its unfinished edge is x = 0 and one short end, the **joint end**, butts against A's front edge. Top B is top A turned 90°, never mirrored.

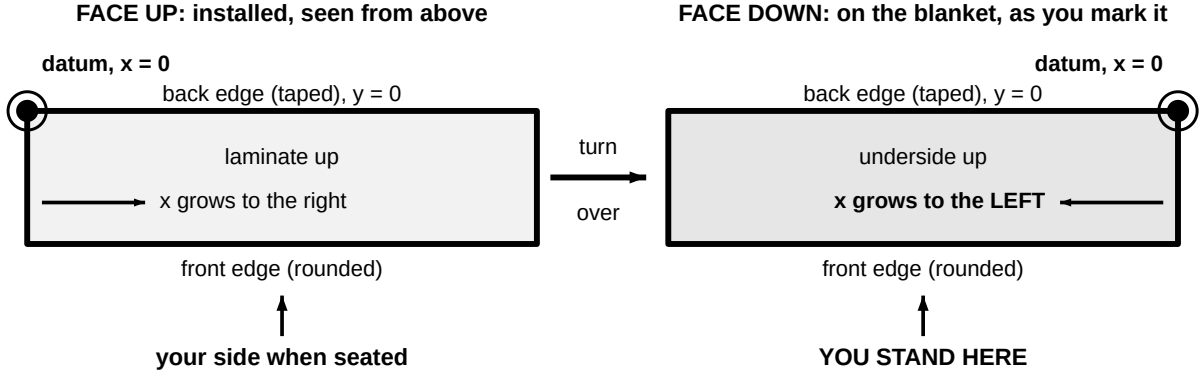
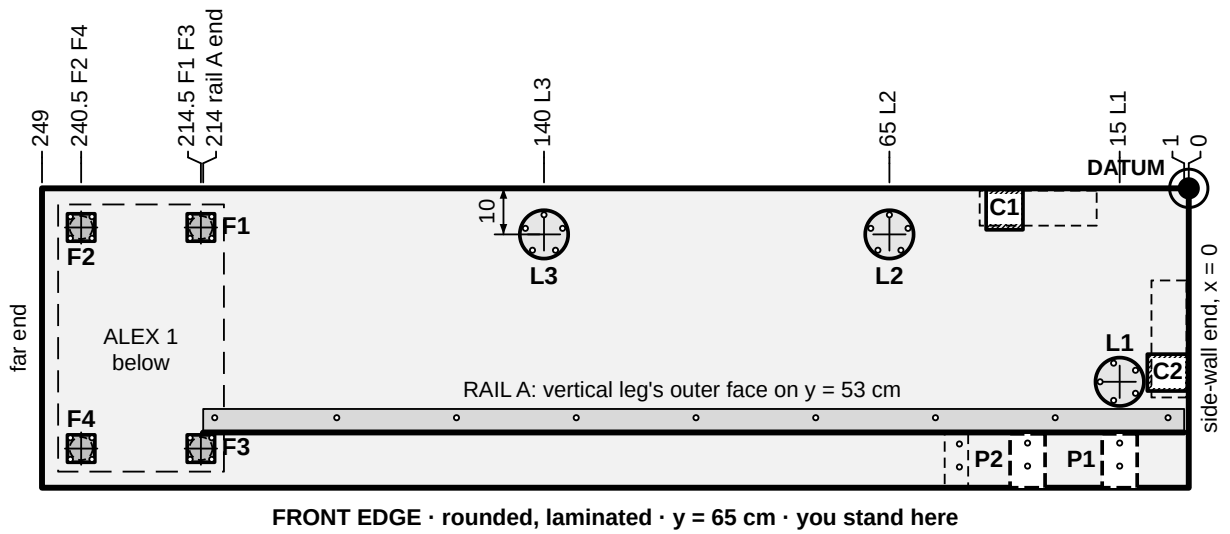


Fig. D-2 · Face-down convention used on every marking drawing. Stand opposite the taped reference edge (top A: at its front edge; top B: at its far end, joint end away from you): the underside is the plan mirrored left to right, so **x grows to the LEFT and the datum side (x = 0) is on your RIGHT**. Top B uses the same mapping.

Top A, face down: overview

1

Calipers on top A: 28 mm **MEASURE ON HARDWARE**. If it is 27 mm, shorten every screw into it by 1 mm. Tape the **unfinished long edge** ("A · BACK EDGE") and the short end you choose as the **side-wall end** ("A · SIDE-WALL END"). Lay top A face down on the blanket and stand at the front edge, back edge away from you.



Scale 1:16. y: tape hooked on the back edge. Details: Figs D-4, D-5.

Fig. D-3 · Top A underside as you see it, all cm. Datum at the far right corner. Circles: OLOV plates (P05); squares: feet plates (P09); grey band: rail A (P06); hatched C1, C2: clamp zones; dashed: pad plates, joining plates P1, P2 (P11), rail B's corner end and ALEX 1 (reference).

Mark	Hook the tape on	Tape reads (cm)	Then
L1	the back edge, tape along the side-wall end	**42**	leg jig, **150** pattern, lip over the side-wall end. Plate ends 6.5 mm short of rail A
L2	the side-wall end, tape along the back edge	**65**	leg jig, 100 pattern, lip over the back edge
L3	the side-wall end, tape along the back edge	**140**	leg jig, 100 pattern, lip over the back edge
F1	the side-wall end, tape along the back edge	**214.5**	T-square, lip over the back edge, hole 85 and its four plate holes
F3	the side-wall end, tape along the **front** edge	**214.5**	T-square, lip over the front (rounded) edge, hole 85 and its four plate holes
F2, **F4**	no tape: feet corner jig on the far back and far front corners		8.5 cm in from both edges
Rail A line	the back edge, at both ends of the line and the middle	**53**	line from 1 to 214 on the side-wall-end tape; the vertical leg's outer face goes on it (12 cm from the front edge)
C1	the side-wall end, along the back edge	**36 to 44**	keep clear, 9 cm in from the back edge
C2	the back edge, along the side-wall end	**36 to 44**	keep clear, 9 cm in from the side-wall end
Pad plate 1	the side-wall end, along the back edge	**20.0 to 45.4**	outline only, 0.5 to 8.1 cm in from the back edge
Pad plate 2	the back edge, along the side-wall end	**20.0 to 45.4**	outline only, 0.5 to 8.1 cm in from the side-wall end
P1 / **P2**	the side-wall end, along the front edge	**11.2–18.8 / 31.2–38.8**	outline only, from 53.3 cm (back-edge tape) to the front edge

All positions **SOURCE-DEFINED** (docs/build-manual.md step 2). Pilots: 1/8 in, started 3–4 mm through the jig; finishing them to depth and fitting the legs and feet follows in the next chapter.

Top A: the corner

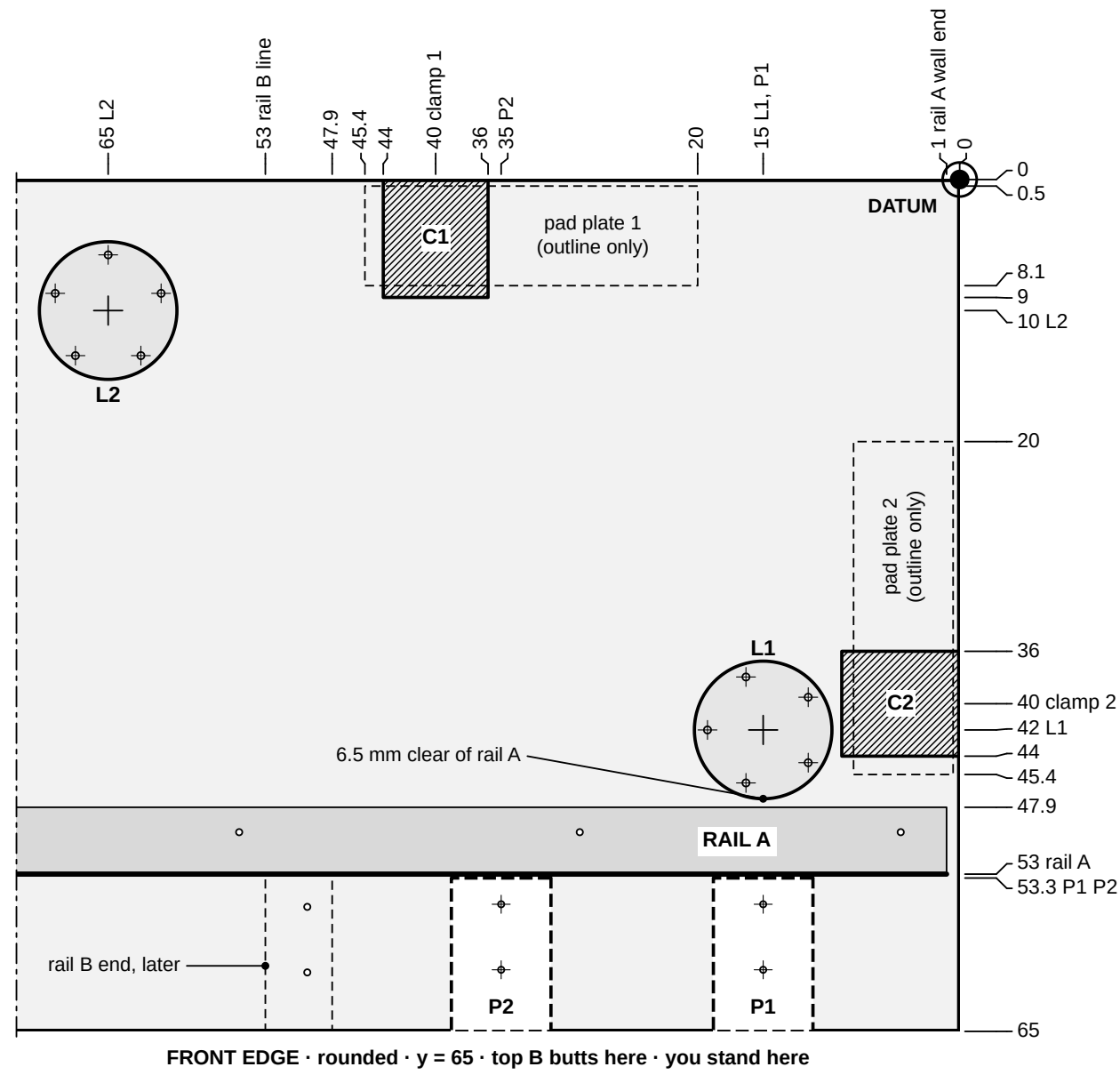
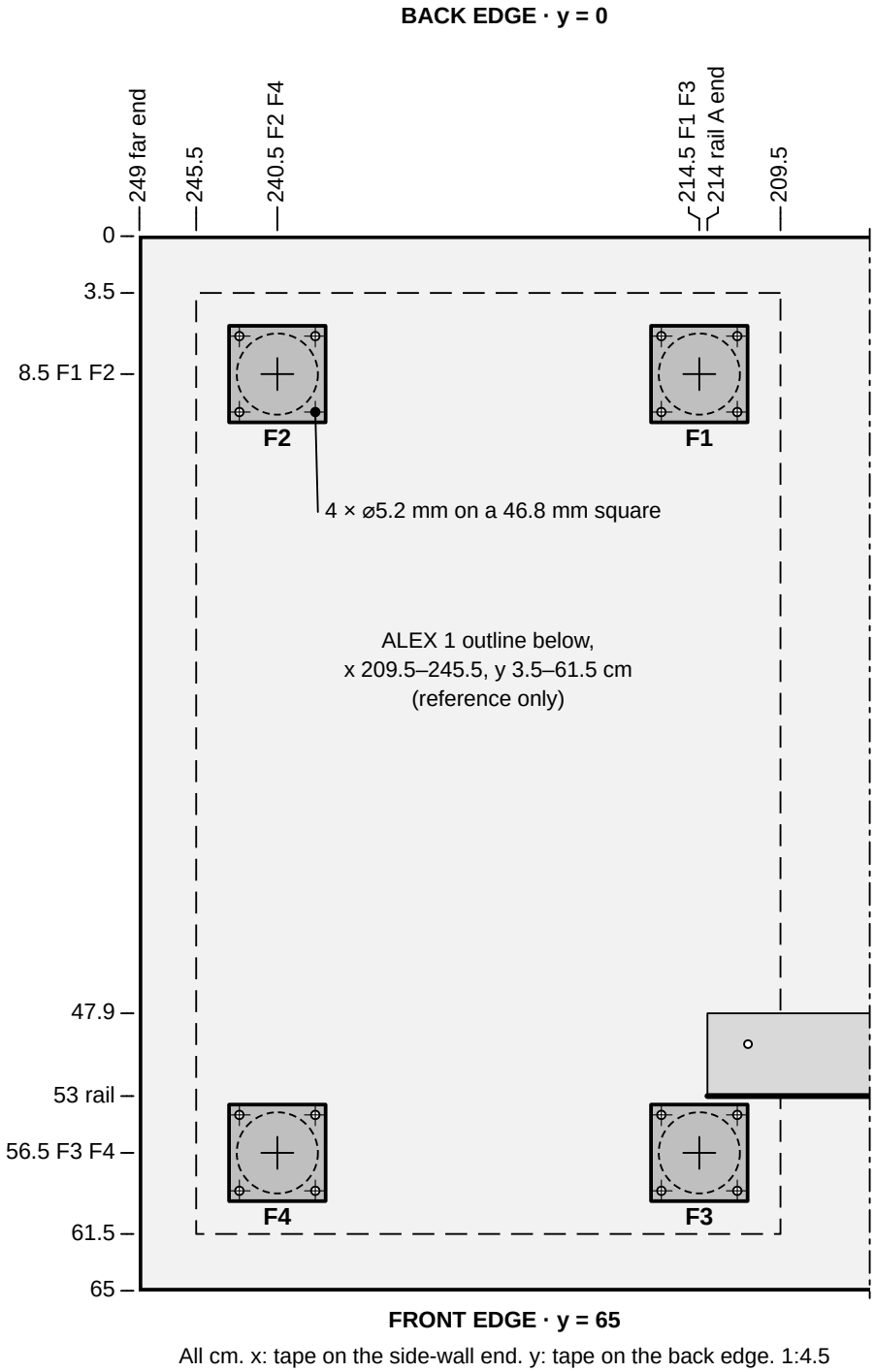


Fig. D-4 · Top A corner, face down, 1:5, all cm. The numbers along the top are the tape hooked on the side-wall end; the numbers on the right are the tape hooked on the back edge.

2 **L1:** hook the tape on the back edge and tick **42 cm** along the side-wall end. Hook the leg jig's lip over the **side-wall end** at that tick and use its **150** pattern (dashed circle, ringed holes): start the five 1/8 in pilots. **L2:** tape on the side-wall end, tick **65 cm** along the back edge; lip over the back edge, **100** pattern (solid circle).

3 Rail A line: tape hooked on the back edge, tick **53 cm** near each end and in the middle; join the ticks from **1 cm** to **214 cm** (side-wall-end tape). Hatch C1 and C2 with pencil. Draw the two pad-plate outlines and the P1, P2 outlines. Nothing is drilled inside C1, C2 or the outlines.

Top A: the far end (feet for ALEX 1)



All cm. x: tape on the side-wall end. y: tape on the back edge. 1:4.5

Fig. D-5 · Top A far end, face down, 1:4.5, all cm.

4 **F2, F4:** no tape. Drop the feet corner jig over the far back corner, then the far front corner, fences against both edges. Start the four 1/8 in pilots through it.

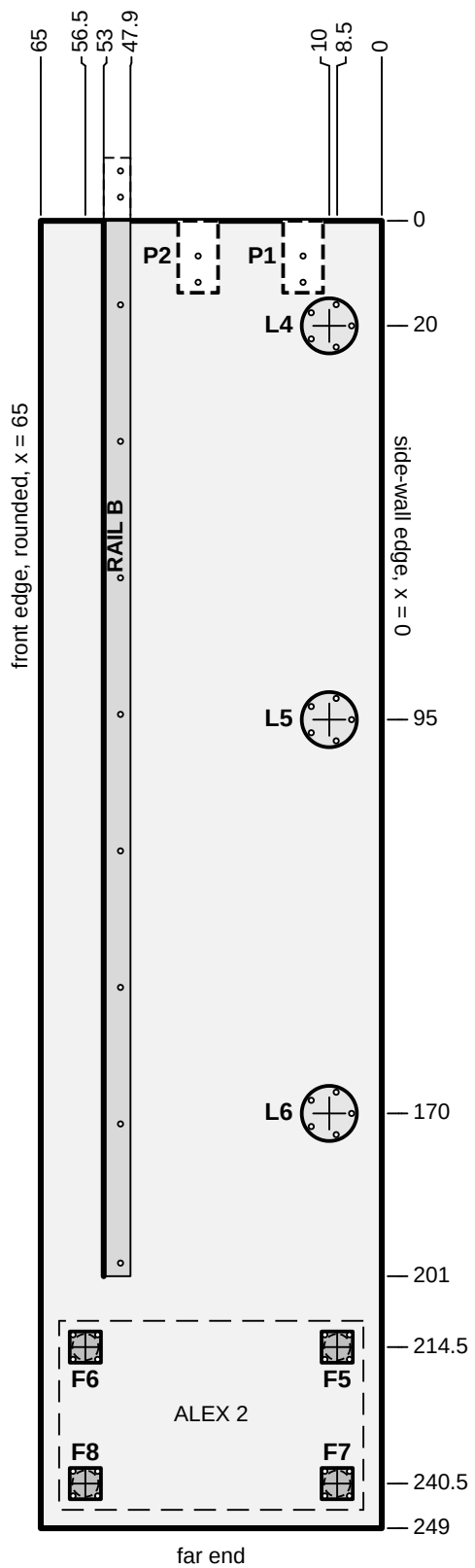
5 **F1:** tape hooked on the side-wall end reads **214.5 cm** along the back edge: tick it. T-square lip over the back edge on the tick, scriber through hole 85 to check, then start the four pilots round it. **F3:** the same, **214.5 cm** along the **front** edge, lip over the front edge.

6 **L3:** tape on the side-wall end reads **140 cm** along the back edge; leg jig, 100 pattern (Fig. D-3).

Feet plate: 59.5 mm square, four ⌀5.2 mm holes 46.8 mm apart **VERIFIED**. Plates square with the top's edges.

The ALEX 1 outline is for orientation only; nothing is marked for it.

Top B, face down: overview



All cm. Right: tape hooked on the joint end.
Top: tape on the side-wall edge. 1:14

Fig. D-6 · Top B underside, joint end away from you, 1:14, all cm.

7

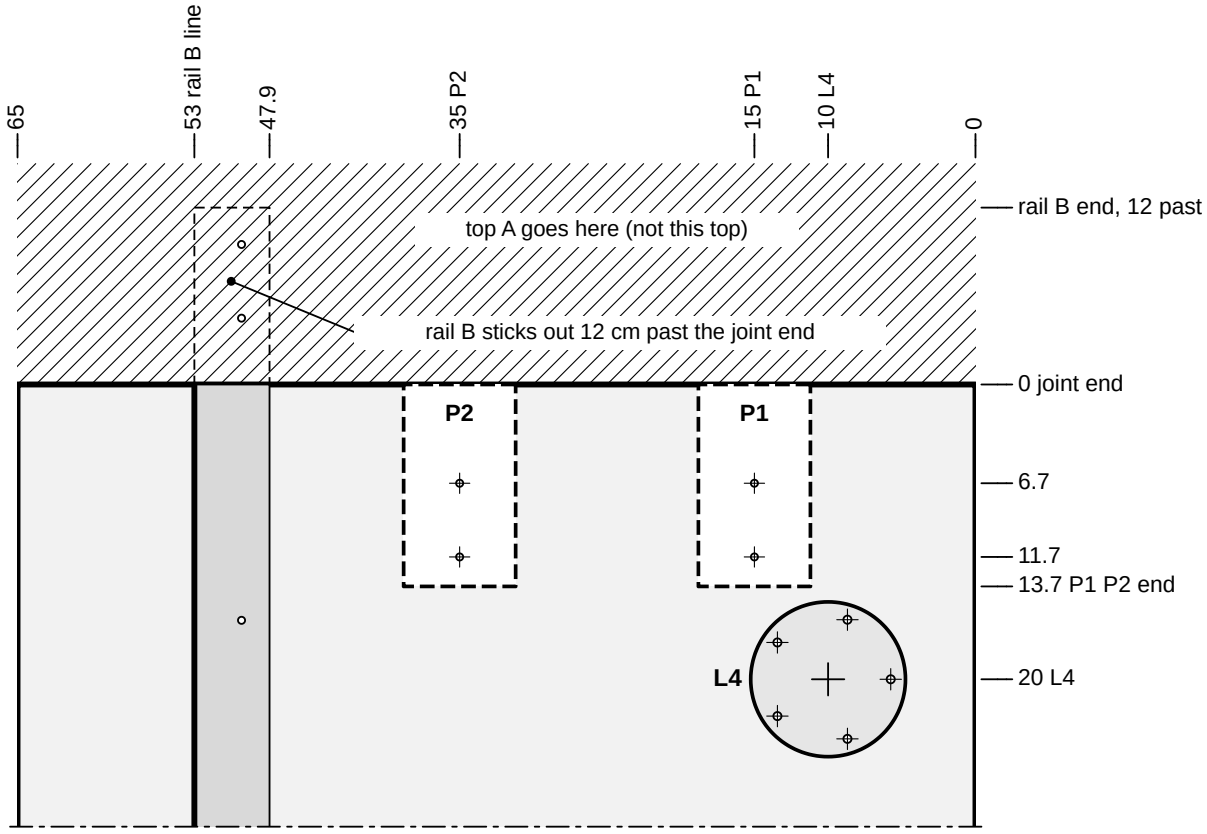
Calipers on top B: 28 mm MEASURE ON HARDWARE. Tape its **unfinished long edge** ("B · SIDE-WALL EDGE") and the short end that will butt against top A ("B · JOINT END"). Lay it face down and stand at the far end, joint end away from you: the side-wall edge is on your right.

> On top B every station along the length is read with the tape **hooked on the joint end**. The numbers below are what the tape shows.

Mark	Tape on the joint end reads (cm)	Then
L4	**20**	along the side-wall edge; leg jig, 100 pattern, lip over the side-wall edge
L5	**95**	
L6	**170**	
F5	**214.5**	along the side-wall edge; T-square hole 85, lip over the side-wall edge
F6	**214.5**	along the **front** edge; T-square hole 85, lip over the front edge
F7, **F8**	no tape	feet corner jig on the two far corners
Rail B line	**0 to 201**	at 53 cm from the side-wall edge (12 cm from the front edge); the vertical leg's outer face goes on it; the rail's corner end sticks out 12 cm past the joint end
P1 / **P2**	**0 to 13.7**	outline only, 11.2–18.8 / 31.2–38.8 cm from the side-wall edge; screws later at 6.7 and 11.7 cm

All positions SOURCE-DEFINED (docs/build-manual.md step 3). L4–L6 have one pilot pointing at the side wall, like L2 and L3.

Top B: the joint end



All cm. Top: tape hooked on the side-wall edge. Right: tape hooked on the joint end.
Joint end away from you, x grows to the left. Scale 1:5

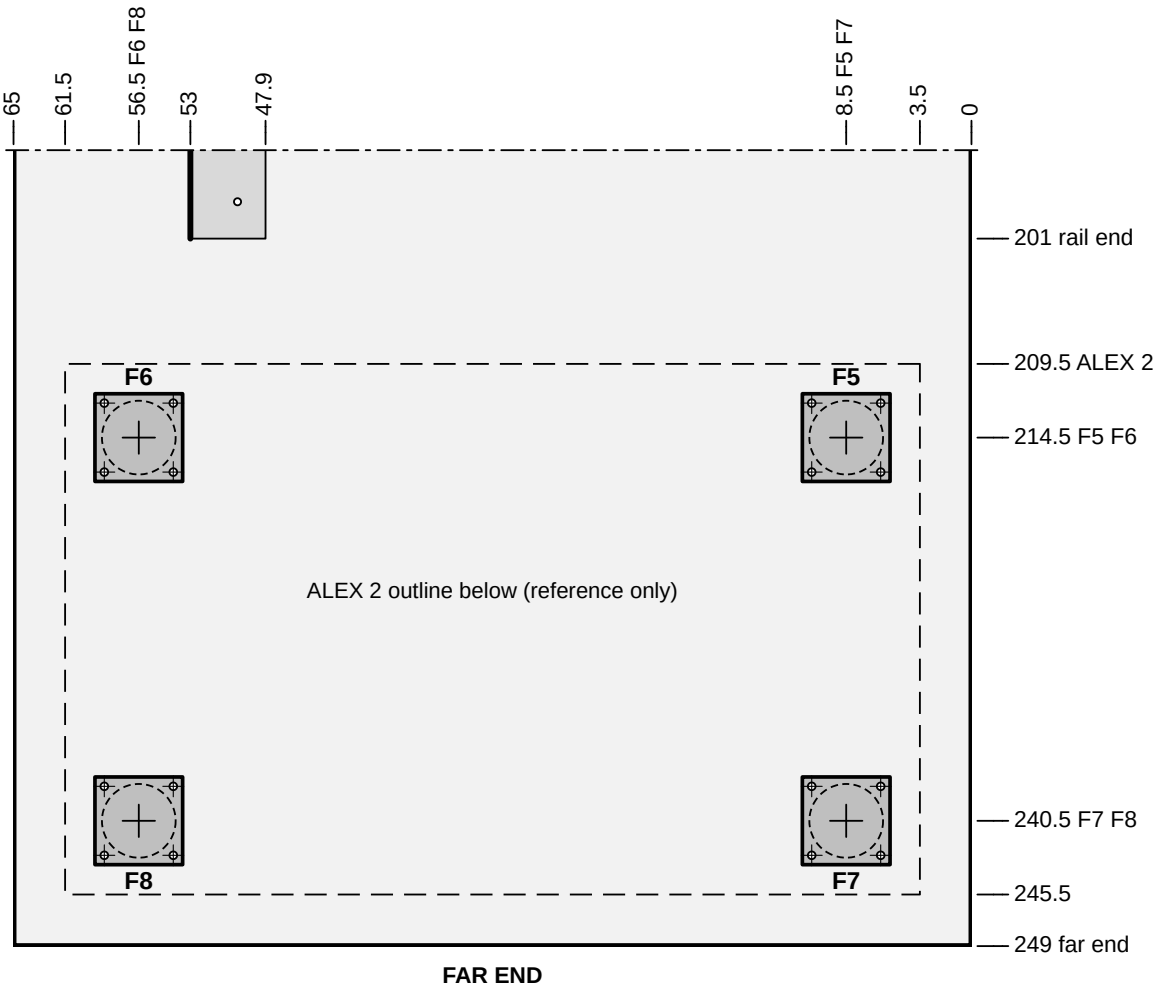
Fig. D-7 · Top B joint end, face down, 1:5, all cm. The hatched area beyond the joint end is where top A will be; rail B's corner end lies there.

8

L4: tape hooked on the joint end reads **20 cm** along the side-wall edge; leg jig, 100 pattern, lip over the side-wall edge. Draw the **rail B line**: tape hooked on the side-wall edge, tick **53 cm** near each end and in the middle; join the ticks from the joint end to **201 cm** on the joint-end tape. Draw the P1 and P2 outlines from the joint end to **13.7 cm**.

Rail B's holes 1 and 2 (shown beyond the joint end) go into top A when the tops are joined; on top B only its holes 3 to 10 are used.

Top B: the far end (feet for ALEX 2)



All cm. Top: tape on the side-wall edge. Right: tape hooked on the joint end. 1:5

Fig. D-8 · Top B far end, face down, 1:5, all cm.

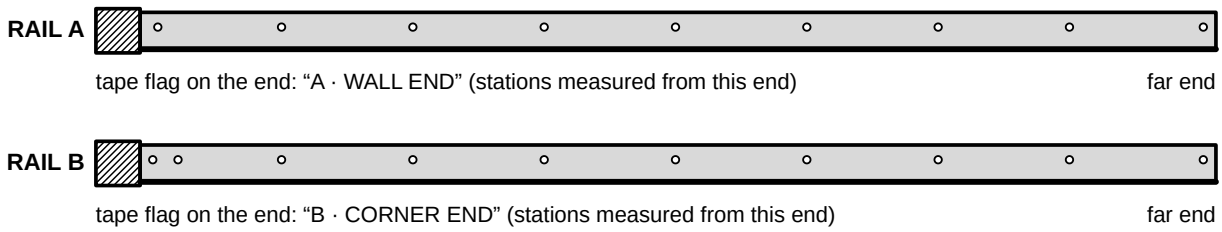
9

F7, F8: no tape: feet corner jig over the two far corners, fences against both edges; start the four 1/8 in pilots. **F5:** tape hooked on the joint end reads **214.5 cm** along the side-wall edge: T-square, lip over the side-wall edge. **F6: 214.5 cm** along the front edge, lip over the front edge. **L5, L6:** leg jig, 100 pattern, at **95 cm** and **170 cm** on the joint-end tape (Fig. D-6).

Rails: labels and orientation

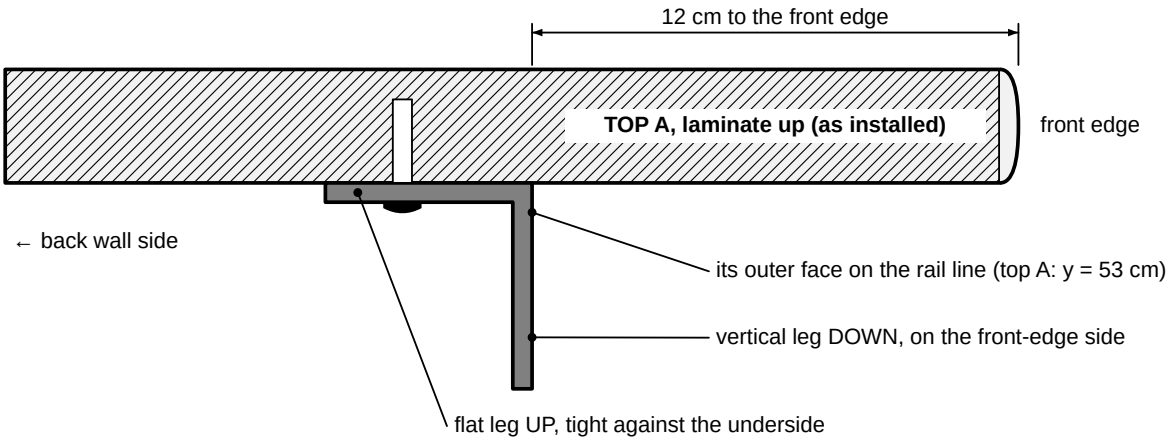
10

The two steel angles arrive identical; either can be rail A. Tape one end of each: **"A · WALL END"** and **"B · CORNER END"**. Every station on a rail is measured from its taped end. Measure both delivered lengths **MEASURE ON HARDWARE** (213 cm layout length; the far end may carry the extra).



Bench view, flat leg up. Holes shown for orientation; positions are in chapter E.

Fig. D-9 · The two rails on the bench, flat legs up, labelled ends left.



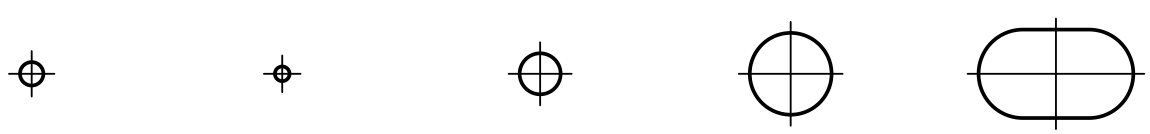
Section through top A and rail A, end-on, about 1:2. Rail B on top B: the same, outer face on x = 53 cm.

Fig. D-10 · How a rail sits under its top: **flat leg UP** against the underside, all screws through it; **vertical leg DOWN** on the front-edge side, its outer face on the rail line, 12 cm from the front edge. Rail A: outer face at 53 cm from the back edge, wall end 1 cm in from the side-wall end. Rail B: outer face at 53 cm from the side-wall edge, corner end 12 cm past top B's joint end.

The printed rail end caps are handed: cap A on rail A's far end and rail B's far end, cap B on rail A's wall end; rail B's corner end takes no cap.

Hole types and depths

The three hole types



pilot ⌀3.2 (1/8 in) into the tops: legs, feet, rails, joining plates

pilot ⌀2.0 (5/64 in) into the ALEX tops and the trim edge

clearance ⌀5.56 (7/32 in): rail flat legs, for #10 screws

through ⌀11.1 (7/16 in) rail vertical legs, 6 holes, 3/8 bolts

slot 21 × 12 in each fitting leg, bought, not drilled

Fig. D-11 · Diameters in mm, 1:1. Pilot holes go into wood; clearance and through holes go through steel. The fittings come with 21 × 12 slots (not drilled); the slot plug centres the rail's 7/16 in pilot in them.

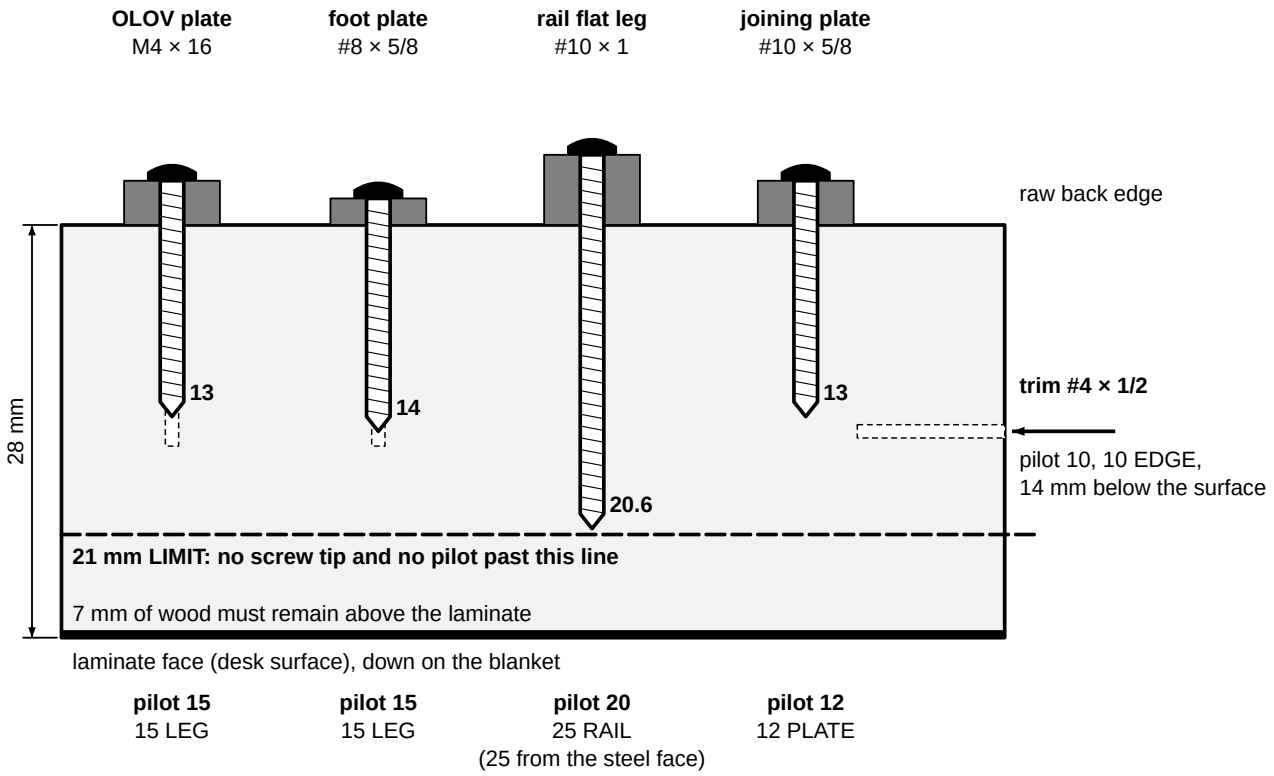
Every fastener, its pilot and its puck

Fastener	Qty	Goes through	Into the wood (mm)	Pilot	Puck
H01 · M4 × 16 (with the OLOV)	30	3 mm leg plate	13	1/8 in, 15 mm deep	15 LEG
H02 · #8 × 5/8 in pan head	32	feet plate (1.5–2 mm)	14	1/8 in, 15 mm deep	15 LEG
H03 · #10 × 1 in pan head, Robertson	19	4.8 mm angle	20.6	1/8 in: the 25 RAIL puck stands on the steel and stops the bit 20 mm into the wood	25 RAIL
H04 + H05 · #10 × 5/8 in with washer	8	3 mm joining plate	13	1/8 in, 12 mm deep	12 PLATE
H06 · #6 × 5/8 in flat head	24	2.5 mm ring flange	13, into the ALEX top	5/64 in, 12 mm deep	12 ALEX
H07 · #4 × 1/2 in flat head (optional trim)	48	trim lip	sideways into the raw back edge, 14 mm below the surface	5/64 in, 10 mm deep	10 EDGE
H08 + H09 + H10 · 3/8-16 bolts	6	fittings and rail vertical legs	none	7/16 in, bench-drilled	—

Depths SOURCE-DEFINED (guide.html fastener table, facts.json). The bolt holes and the 7/32 in screw holes in the steel are drilled on the bench (chapter E); everything into wood goes through a jig first and is finished with its puck.

Never through the laminate

> **NEVER THROUGH THE LAMINATE**
> No screw tip and no pilot goes deeper than **21 mm** into a 28 mm top: at least **7 mm** of wood must remain under the laminate face. Set every pilot depth with the drill guide's puck, never by eye. If a top measures 27 mm, shorten every screw into it by 1 mm.



Section through a 28 mm top, face down as you drill it (underside up), 2:1. Grey: the part being screwed on. Dashed: the pilot hole. Bold numbers: how far the screw goes into the wood, mm.

Fig. D-12 · Every fastener into the tops at its depth, 2:1, mm.

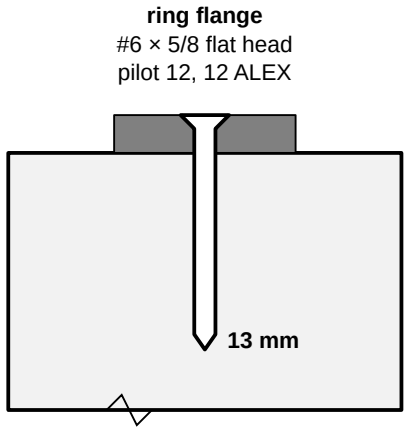


Fig. D-13 · Ring screws go into the ALEX top, not the desk. **MEASURE ON HARDWARE** panel thickness.

Setting a puck

11 Loosen the chuck. Drop the bit tip-first into the SET tower's hole for that bit (45 mm for 1/8 in, 35 mm for 5/64 in), rest the chuck nose on the tower, tighten. Re-set every time you change bits.

12 Stand the puck flat on the surface (on the rail's flat leg for 25 RAIL), bit down its bore, drill until the chuck nose lands on the puck. The hole is then the depth written on the puck.

If your chuck nose is domed, drill one test hole in scrap and measure it first.